REDBIRD FLIGHT SIMULATIONS

QUALIFICATION AND APPROVAL GUIDE (QAG)

C510 QUALIFICATION AND APPROVAL GUIDE
VERSION 2.0A



ADVANCED AVIATION TRAINING DEVICE

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FAA APPROVED QAG
Signature and Date

June 15, 2018

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LOG OF REVISIONS

Description of Changes	Version	Edited By
Original	1.0	CG
Added Log of Revisions	2.0	WJ
Updated Section 1 Device Overview	2.0	WJ
Moved Figures and Photos to Section 2	2.0	WJ
Updated Panel images Figure 1 and Figure 2	2.0	WJ
Added Visual Displays – Type 2 Figure 6.4	2.0	WJ
Moved Hardware and Software Components to Section 3	2.0	WJ
Added Win10, Visual Display Type 2, Navigator to Table 1	2.0	WJ
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Moved Statement of Compliance to Section 4	2.0	WJ
Updated Display Requirements (1) and (2)	2.0	WJ
Added Performance table to Section 4	2.0	WJ
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Added new configuration for approval	2.0	WJ
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Revised Section 4	2.0	WJ
Added Figures 6.6 and 6.7 to Section 2	2.0	WJ
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Added Optional Control Loading (Dual) rudder pedals to Section 3	2.0	WJ
Updated GLS in Section 7 to “No”	2.0A	WJ
Updated images 6.3 and 6.4 in Section 2	2.0A	WJ
Updated image 6.1 in Section 2	2.0A	WJ
Added image 6.8 in Section 2	2.0A	WJ

SECTION 1: COMPLIANCE STATEMENT

This Qualification and Approval Guide (QAG) includes a detailed description of all components, functions, and capabilities for the Redbird model C510. This includes any optional aircraft configurations with quality color pictures and diagrams. The required information as described in the current advisory circular, AC 61-136A / FAA Approval of Aviation Training Devices (ATD) and Their Use for Training and Experience, required for FAA approval, is contained within this document. This includes listing all the required qualifying items, functions, and capabilities. This QAG is used to clearly describe and verify the functionality of this aviation training device platform confirming its suitability for airman training and experience. A valid FAA Letter of Authorization (LOA) specifying the ATD allowances must also accompany the training device when utilized for airman training or experience requirements as specified in 14 CFR §61 or 141.

This ATD must continue to maintain its performance and function without degradation. The operator of this device is expected to maintain its condition and functionally when used for airmen training or experience requirements. Only the configurations approved for this model, as described within this QAG, can be utilized when satisfying FAA experience or training requirements. Any additions, changes, or revisions to the model or to the available configurations described within this document must be approved in writing by The General Aviation and Commercial Division, AFS-800.

Any questions concerning FAA approval of ATDs should be directed to the General Aviation and Commercial Division, AFS-800.

SECTION 2: AVIATION TRAINING DEVICE (ATD) DESCRIPTION AND PICTURES

The Redbird model C510 is based on the dimensions and layout of a Cessna Citation Mustang (CE-510) aircraft. This closely represents the overall functionality, performance, avionics, and instrumentation. The platform consists of a cockpit section, instructor's control station, visual display system and an audio system. It incorporates a combination of hardware and software components that is assembled and checked by Redbird Flight Simulations. All hardware elements are permanently installed and designed so the cockpit has the appearance and feel of an actual aircraft. From the pilot's seated position, there are no computer hardware elements such as keyboards, pointing devices, etc. for his or her use.

The C510 provides realistic and true-to-scale cockpit design, avionics, and reliable hardware/software performance. This platform also provides an effective training environment for student and certificated pilots. This includes the capability of practicing scenario based flight training events, simulated equipment failure and emergency procedures, pilot evaluations, instrument procedures/ experience, and facilitating increased pilot proficiency overall.

The Redbird C510 is a versatile and affordable device that has been designed to represent a Cessna Citation Mustang (CE-510). It is equipped with the following notable features:

- 3-axis electric motion platform providing pitch, roll and yaw motions
- Wrap-around exterior visuals provided by 6 (8 optional) LCD screens
- Dual pilot controls including a 2-axis control-loaded yoke and interconnected rudder pedals
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to match the aircraft wherever possible
- An interchangeable instrument panel to allow development of future configurations
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructors station, allowing the instructor to operate from inside or outside the simulator
- Optional supplemental oxygen system for the pilot and/or co-pilot

Configuration Components

Instructor's Station

The Redbird Instructor Station interface is operated through any PC and/or web browser enabled device. The Instructor Station software is available as an installed program on the PC and web-browser based application. PC based Instructor Station can be on either a desktop computer or laptop/mobile PC.



Figure 6.1 Instructors PC/Tablet or Mobile Device

Aircraft Instrument Configurations

The Redbird C510 simulator represents a twin engine turbofan, land, retractable gear, low wing, glass-cockpit airplane representative of a Cessna Citation Mustang, equipped with a three-screen G1000 avionics package.



Figure 6.2 Assembled Cockpit



Figure 6.6 Assembled Cockpit



Figure 6.7 Assembled Cockpit

Controls



Figure 2.1 Pilot Yoke

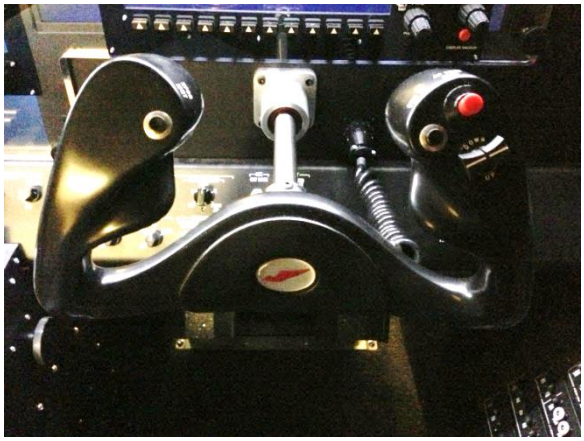


Figure 2.6 Co-Pilot Yoke



Figure 2.2 Pilot Yoke Switches – Left



Figure 2.3 Pilot Yoke Switches – Left



Figure 2.4 Pilot Yoke Switches – Right



Figure 2.7 Co-Pilot Yoke Switches – Left



Figure 2.8 Co-Pilot Yoke Switches – Left



Figure 2.9 Co-Pilot Yoke Switches – Right



Figure 3.1 Throttle Quadrant



Figure 5.1 FMS Keypad



Figure 5.2: Circuit Breaker Panel Pilots Side



Figure 5.3: Circuit Breaker Panel Co-Pilots Side



Figure 5.4: Autopilot Control Panel



Figure 5.5: Lower Switch Panel Pilots Side 1



Figure 5.6: Lower Switch Panel Pilots Side 2



Figure 5.8: Lower Switch Panel Co-Pilots Side



Figure 5.10: Emergency Gear Extension Handle



Figure 5.9: Gear Handle



Figure 5.7: Lower Switch Panel Center



Figure 6.5: Motion Platform and Cockpit Enclosure



Figure 6.8: Pilot & Co-pilot Seats

SECTION 3: TRAINING DEVICE COMPONENTS LIST

Qty	Type	Manufacturer	Name	Description/Function	Configuration
1	Software	Microsoft	Windows 7 or 10	Operating system. (depending on serial number)	All
1	Software	Lockheed Martin	Prepar3D	Simulation engine. (depending on serial number)	All
1	Software	Redbird Flight Simulations	RB Sim	Simulation control and component integration. (depending on serial number)	All
1	Software	Redbird Flight Simulations	FMX Control	Motion system control and component integration.	All – Except when configured as an SD.
1	Software	Redbird Flight Simulations	Instructors Station	Environmental, location and failure controls with map, track and glideslope display. (depending on serial number)	All
1	Software	Redbird Flight Simulations	Navigator	Simulation control and component integration. Environmental, location and failure controls with map, track and glideslope display. (depending on serial number)	All
1	Software	Redbird Flight Simulations	Parrot and Cygnus	Optional software providing ATC and location services	All
1	Software	Mindstar Aviation	Redbird G1000 (C510) and C510 Gauge	Virtual flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	See Configuration
1	Software	Redbird Flight Simulations	Miscellaneous Gauges	Virtual Miscellaneous Gauges	All
1	Software	Mindstar Aviation	Miscellaneous Gauges	Virtual Miscellaneous Gauges	All
1	Software	Flight 1	Compass	Virtual Compass	All
1	Hardware	Redbird Flight Simulations	Simulation Computer	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	All
1	Hardware	Redbird Flight Simulations	Motion Control Computer	Host computer for FMX Control.	All – Except when configured as an SD.
1	Hardware	Industry Standard	Instructors Station Computer or Wireless Mobile Device	Host computer for Instructors Station.	All
1	Hardware	Redbird Flight Simulations	Motion Platform: Type 1	Gimbaled, steel motion platform with movement in pitch, roll and yaw. Includes all motors, sensors and safety controls.	All – Except when configured as an SD.
1	Hardware	Redbird Flight Simulations	Cockpit Enclosure	Cockpit enclosure to exclude distractions.	All
6	Hardware	22” LCD Industry Standard Monitor	Visual Display - Type 1	Flat Panel displays for exterior views.	All – Varies by configuration. See Figure 6.3
8	Hardware	22” LCD Industry Standard Monitor	Visual Display - Type 2	Optional Flat Panel displays for exterior views.	All – Varies by configuration. See Figure 6.4
3	Hardware	19” LCD Industry Standard Monitor	LCD	Flat Panel displays for virtual instruments.	All
1	Hardware	Redbird Flight Simulations	Yoke (Dual)	Dual, Control loaded pitch and roll controller with switches and buttons for airplane systems operation.	All

1	Hardware	Redbird Flight Simulations	Throttle Quadrant	Twin, turbine engine controls with throttle levers, flap position lever and pitch, elevator, aileron and rudder trim controls with indicators for each, FMS Keypad and Emergency gear extension handle.	All
1	Hardware	Redbird Flight Simulations	Rudder Pedals (Dual)	Pilot and co-pilot rudder control pedals with toe brakes.	All – Varies by configuration
1	Hardware	Redbird Flight Simulations	Rudder Pedals (Dual) – Control Loading	Optional Control loading pilot and co-pilot rudder control pedals with toe brakes.	All – Varies by configuration
1	Hardware	Redbird Flight Simulations	Switch Panel	Lower switch panel with airplane configuration and systems controls.	All
1	Hardware	Redbird Flight Simulations	Left Switch Panel	Left side switch panel with airplane systems	All
1	Hardware	Redbird Flight Simulations	Oxygen System	Optional oxygen system for the pilot and/or copilot	All
1	Hardware	Redbird Flight Simulations	Circuit Breaker Panel	Pilot side circuit breaker panel.	All
1	Hardware	Redbird Flight Simulations	Overhead Control Panel	Overhead control panel.	All
1	Hardware	Redbird Flight Simulations	Instrument Controls Overlay	Flight instruments, radios, airplane configuration and systems controls as required for each configuration.	See Configuration

Table 1: Training Device Component List (Above)

Statements of Compatibility of Software and Hardware

AC 61-136A, Appendix 1, 5.b.:

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Microsoft Windows (Operating System)

This is to certify that Microsoft Corporation, the owner and developer of the Windows operating system, has evaluated that their operating system works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

Lockheed Martin Prepar3D (Simulation Engine)

This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software

Mindstar Aviation Redbird G1000 (C510) and C510 Gauge

This is to certify that Mindstar Aviation, the owner and developer of the Redbird G1000 (C510) and C510 Gauge, and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Mindstar Aviation software package is fully compatible with the Redbird Flight Simulations C510 and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations.

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Flight 1 Compass

This is to certify that Flight 1, the owner and developer of the Compass and Redbird Flight Simulations, the hardware manufacturer and component integrator, have demonstrated that the Compass software is fully compatible with the Redbird Flight Simulations model C510. Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations.

SECTION 4: AVIATION TRAINING DEVICE (ATD) DESIGN CRITERIA LIST

The following section provides the detailed "word for word" listing and design criteria of each of the required items, functions, and capabilities (listed in AC 61-136A, for BATD requirements Appendix 2 and the additional AATD items of Appendix 3) and operational performance value/scale (as applicable) for each of the functions described for the Redbird C510.

Basic ATD Requirements List [Appendix 2 items]

All configurations for this model, as noted, meet AC 61-136A, Appendix 2 requirements

a. The Redbird C510 meets the following Control Input requirements:

(1) Physical flight and aircraft control systems are designed such that they are recognizable as to their function and how they are manipulated solely from their appearance. No keyboard, mouse, or gaming joystick is used to control the aircraft.
(2) Virtual controls are limited to setting the aircraft configuration, location, weather conditions, and pausing action.
(3) No keyboard or mouse is used to set or position any of the below listed features of this device. The required and additional equipment is operated in the same manner in which it would be operated in the aircraft represented (as applicable). This includes the landing gear, wing flaps, cowl flaps, carburetor heat, and mixture, propeller, and throttle controls.
(4) The physical arrangement, appearance, and operation of controls, instruments, and switches (as applicable) closely represents the model of aircraft including: Master/battery, magnetos, alternators/generators, auxiliary power units, fuel boost/prime pumps, avionics master, pitot heat, rotation beacon/strobe, navigation, taxi, and landing lights.
(5) Only the software evaluated and approved by the FAA is installed on this computer system.

b. The Redbird C510 meets the following Control Requirements:

(1) Airplane
  (a) Self-centering control yoke or control stick that allows continuous adjustment of pitch and bank
  (b) Self-centering rudder pedals rudder pedals that allow continuous adjustment of yaw and corresponding reaction in heading and roll.
  (c) Throttle or Power controls/levers as applicable that allows for continuous movement from idle to full power, with corresponding changes in pitch and yaw.
  (d) Mixture/condition, propeller, and throttle/power control(s) as applicable to the model aircraft represented
  (e) Controls for the following items applicable to the aircraft represented:
    - Wing flaps,
    - Pitch trim,
    - Communication and navigation radios,
    - Clock or timer,
    - Gear handle (if applicable),
    - Transponder,
    - Altimeter,
    - Carburetor heat (if applicable), and
    - Cowl flaps (if applicable).

c. The Redbird C510 meets the following Control Input requirements:

(1) Time from control input to recognizable system response must be without delay (i.e., not appear to lag in any way). “Redbird Flight Simulations, Inc.” verifies that the “Redbird C510” meets this requirement.

(2) The control inputs are tested by the computer and software at each session startup and displayed as a confirmation message or a “warning message” that the transport delay time or any design parameter is out of original tolerances. It is not possible to continue the training session unless the problem is resolved and all components are functioning properly. This test considers the items listed in the display requirements.

d. The Redbird C510 meets the following Display Requirements:

(1) **Instruments and indicators** replicated and properly located as appropriate to the aircraft model represented:

Flight instruments in a standard configuration, represented as traditional “round dial” flight instruments, or as an electronic primary flight instrument display (PFD) and multi-function display (MFD) with reversionary and back-up flight instruments.

- A sensitive **altimeter** with incremental markings each 20 feet or less, operable throughout the normal operating range of the aircraft or family of aircraft represented.
- A **magnetic direction indicator**.
- A **heading indicator** with incremental markings each 5 degrees or less, displayed on a 360 degree circle. Arc segments of less than 360 degrees may be selectively displayed if desired or required, as applicable to the aircraft or family of aircraft represented.
- An **airspeed indicator** with incremental markings as shown on the aircraft or family of aircraft represented; airspeed markings of less than 40 knots need not be displayed.
- A **vertical speed indicator** with incremental markings each 100 feet per minute (fpm) for both climb and descent, for the first 1,000 fpm of climb and descent, and at each 500 fpm climb and descent for the remainder of a minimum ± 2,000 fpm total display, or as applicable to the aircraft or family of aircraft being represented.
- A **gyroscopic rate-of-turn indicator** or equivalent with appropriate markings for a rate of 3 degrees per second turn for left and right turns. If a turn and bank indicator is used, the 3 degrees per second rate index must be inside of the maximum deflection of the indicator.
- A **slip and skid indicator** with coordination information displayed in the conventional skid ball format where a coordinated flight condition is indicated with the ball in the center position. A bar or split image triangle indication may be used if the aircraft is equipped with an electronic PFD.
- An **attitude indicator** with incremental markings each 5 degrees of pitch or less, from 20 degree pitch up to 40 degree pitch down or as applicable to the aircraft or family of aircraft represented. Bank angles must be identified at “wings level” and at 10, 20, 30, and 60 degrees of bank (with an optional additional identification at 45 degrees) in left and right banks.
- **Engine instruments** as applicable to the aircraft or family of aircraft being represented, providing markings for normal ranges and minimum and maximum limits.
- A **suction gauge** or instrument pressure gauge, if applicable, with a display appropriate to the model of aircraft represented.
- A **flap setting indicator** that displays the current flap setting. Setting indications should be typical of that found in an actual aircraft.
- A **pitch trim indicator** with a display that shows zero trim and appropriate indices of airplane nose down and airplane nose up trim, as would be found in an aircraft.
- **Communication radio(s)** with display(s) of the radio frequency in use.
- A **clock** with incremental markings for each minute and second, or a timer with a display of minutes and seconds,

- A **transponder** that displays the current transponder setting.
- A **fuel quantity indicator(s)** that displays the fuel remaining, either in analog or digital format, as appropriate for the aircraft or family of aircraft represented.

Navigation Equipment: Replicates both precision and non-precision instrument navigation, including approach procedures (with an aural identification feature if normally available), and a marker beacon receiver. Examples: an ILS receiver, non-directional radio beacon (NDB), Global Positioning System (GPS), Localizer (LOC) or Very high frequency Omnidirectional Range (VOR). Graduated markings as indicated below are present on each course deviation indicator as applicable.

Required markings include:

- One-half dot or less for **course/glide slope deviation indicator** (i.e., VOR, LOC, or ILS),
- 5 degrees or less for bearing deviation for automatic direction finder (ADF) and radio magnetic indicator (RMI), as applicable.
- If equipped with a PFD and/or MFD, the flight and navigation information and guidance must replicate the avionics manufactures same scales and navigation information presentation provided in the aircraft equipment.

(2) All instrument displays listed above must be visible during all flight operations. The update rate of all displays must provide an image of the instrument that:
   (a) Does not appear to be out of focus or illegible
   (b) Does not appear to "jump" or "step" to a distracting degree during operation.
   (c) Does not appear with distracting jagged lines or edges.
   (d) Does not appear to lag relative to the action and use of the flight controls.

(3) Control inputs are reflected by the flight instruments in real time and without a perceived delay in action. Display updates or actions show all changes (within the total range of the replicated instrument) that are equal to or greater than the following:

   (a) Airspeed indicator: change of 5 knots.
   (b) Attitude indicator: change of 2 degrees in pitch and bank.
   (c) Altimeter: change of 10 feet.
   (d) Turn and bank: change of ¼ standard rate turn.
   (e) Heading indicator: change of 2 degrees.
   (f) Vertical speed indicator (VSI): change of 100 fpm.
   (g) Tachometer: change of 25 rpm or 2 percent of turbine speed.
   (h) VOR/ILS: change of 1 degree for VOR or ¼ of 1 degree for ILS.
   (i) ADF: change of 2 degrees.
   (j) GPS: change as appropriate for the model of GPS based navigator represented.
   (k) Clock or timer: change of 1 second.

(4) **Flight Displays reflect proper dynamic behavior**. Examples: VSI reading of 500 fpm reflects a corresponding movement in altimeter. Throttle movement reflects an appropriate change in pitch, rpm indication, or the power indicator. Pitch changes reflect an appropriate change in airspeed and altitude.

e. The Redbird C510 meets the following Flight Dynamics Requirements:

(1) All flight dynamics are comparable to the way the represented airplane performs and handles.
(2) All simulated aircraft performance parameters (such as maximum speed, cruise speed, stall speed, maximum climb rate and hovering/sideward/forward/rearward flight) are comparable to the airplane

represented. A performance table is included in the QAG for each aircraft configuration for sea level and 5,000 feet.

(3) Vertical lift component changes as a function of bank/load factor, comparable to the aircraft represented.

(4) Changes in flap setting and gear position are accompanied by changes in flight dynamics comparable and reasonably equivalent to the aircraft represented

(5) The presence and intensity of wind and turbulence must be reflected in the handling and performance qualities of the simulated aircraft and should be comparable to the way the aircraft or family of aircraft represented performs and handles.

f. The Redbird C510 meets the following Instructional Management Requirements:

(1) The instructor can pause the system at any point during the training session for the purpose of administering instruction regarding the training task at hand.

(2) If a training session begins with the "aircraft in the air" and ready for the performance of a particular procedural task, the instructor can manipulate the following system parameters independently of the simulation:

- Aircraft geographic location,
- Aircraft heading,
- Aircraft airspeed,
- Aircraft altitude, and
- Wind direction, speed, and turbulence.

(3) The system is capable of recording both a horizontal and vertical track of aircraft movement for the entire training session for later playback and review.

(4) The instructor can disable any of the instruments prior to or during a training session and simulate failure of any of the instruments without stopping the flight simulation to affect the failure.

(5) The ATD has an available navigational database that is local to the training facility. All navigational data must be based on procedures as published per 14 CFR part 97.

Advanced ATD Requirements List [Appendix 3 items]

All configurations, as noted in AC 61-136A, Appendix 3 meet these additional AATD design criteria items listed.

a. The Redbird C510 meets the following additional AATD CRITERIA:

(1) A realistic shrouded (enclosed) or unshrouded (open) cockpit design and instrument panel arrangement representing a specific model aircraft cockpit.

(2) **Cockpit knobs, system controls, switches, and/or switch panels in realistic sizes and design** appropriate to each intended functions, in the proper position and distance from the pilot's seated position, and representative of the category and class of aircraft being represented.

(3) Primary flight and navigation instruments appropriately sized and properly arranged that exhibit neither stepping nor excessive transport delay.

(4) Digital Avionics Panel

(5) **Global Positioning System (GPS)** navigator with moving map display.

(6) **Two-axis autopilot**, and, as appropriate, a flight director (FD). This is not required for an ATD representing a helicopter.

(7) **Pitch trim** (manual or electric pitch trim) permitting indicator movement either electrically or analog in an acceptable trim ratio.
(8) An independent **visual system**, panel, or screen that provides realistic cues in both day and night visual flight rules (VFR) and instrument flight rules (IFR) meteorological conditions to enhance a pilot's visual orientation in the vicinity of an airport including:
- Adjustable visibility parameters; and
- Adjustable ceiling parameters.

(9) A **pilot seat** appropriate to the aircraft configuration.
(10) **Rudder pedals** secured to the cockpit floor structure, or that can be physically secured to the floor beneath the device in proper relation to cockpit orientation.
(11) Push-to-talk switch on the control yoke.
(12) A **separate instructor station** to permit effective interaction without interrupting the flight in overseeing the pilot's horizontal and vertical flight profiles in real time and space. This must include the ability to:
(a) Oversee tracks along airways, holding entries and patterns, and Localizer (LOC) and glideslope (GS) alignment/deviation (or other approaches with a horizontal and vertical track).
(b) Function as air traffic control (ATC) in providing vectors, etc., change in weather conditions, ceilings, visibilities, wind speed and direction, light/moderate/severe turbulence, and icing conditions.
(c) Invoke failures in navigation and instruments, radio receivers, landing gear and flaps, engine power (partial and total), and other aircraft systems (pitot, electric, static, etc.) by using either a keyboard or mouse.
(13) Capable of simulating the emergency procedures for which a checklist is prescribed in the aircraft pilot's operating handbook (POH) or flight manual.

b. The Redbird C510 meets the following additional encouraged AATD CRITERIA:

(2) Motion capability with simulated bank, yaw, and pitch ques (*Requires motion platform)
(3) Loss of performance and aerodynamic changes from ice accretion
(4) Realistic aircraft engine sound appropriate to the aircraft configuration, power settings, and speed.
(5) A magnetic compass with incremental markings each 5 degrees, that displays the proper lead or lag during turns, and displays incremental markings typical of that shown in the aircraft.

SECTION 5: AIRCRAFT CONFIGURATION

List of Previously Approved Configurations (QAG v1.0):

C510



MTOW	FWD/AFT CG Limit
8645 lbs	287.04 in / 292.46 in

- **Yoke – Center (See Figures 2.1-2.8, 6.2)**
- **Throttle – Twin Engine Turbine Throttle (See Figure 3.1)**
- **Glass Cockpit G1000 (3)PFD/MFD & Autopilot**

Additional configurations included in this version (QAG v 2.0)

NONE

Performance Table

Aircraft Model	V_{SO}	V_{S1}	V_X	V_Y	V_{MO}	V_{MCA}	KTAS @ Cruise / 75% power setting	Rate of Climb (fpm) @ (V_Y) / Full Power*	Single Engine Rate of Climb (V_{YSE})
Cessna Mustang (C-510)	91 KIAS	73 KIAS	150 KIAS	170 KIAS	250 KIAS	81* KIAS	218 KTAS**	3010 fpm	870 fpm (4.9 Climb Gradient)
12,000' -->							245 KTAS	2315 fpm	390 fpm (1.9 Climb Gradient)

* 15 degrees flaps (takeoff & approach)

** Performance @ 5,000'

SECTION 6: VISUAL SYSTEM WITH IFR, VFR, DAY, AND NIGHT CAPABILITY

Redbird C510 Visual System
The visual system is capable of providing a field-of-view of a minimum of 45 degrees horizontally and 30 degrees vertically, simultaneously for each pilot, including adjustable cloud base and visibility in night, dusk and day scenes.

- The Redbird C510 provides a means of recording the visual response time for the visual system that is installed.
- The Redbird C510 visual system is free of optical discontinuities and artifacts that create non-realistic cues.
- The visual system is directly displayed on six (6) or (8) LCD monitors inside the cockpit enclosure, situated in an arc around the Pilot. Each monitor is 28 cm tall, and 47 cm wide. Based upon the designated Pilot Eye Point, these monitors provide a horizontal FOV of 220 (6 monitors) to 260 (8 monitors) degrees and a vertical FOV of 30 degrees.

Daylight: The visual system provides full color presentations and sufficient surfaces with appropriate textural cues to conduct a visual approach, landing and airport movement. Surface shading effects are consistent with the simulated sun position.

Twilight: The visual system provides full color presentations of reduced ambient intensity, sufficient surfaces with appropriate textural cues that include self-illuminated objects such as road networks, ramp lighting and airport signage, to conduct a visual approach, landing and airport movement. Scenes include a definable horizon and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by representative ownship lighting.

Night: The visual system provides the same as above except the portrayal of reduced ambient intensity; therefore, there is no ground cues that are not self-illuminating or illuminated by ownship lights.

Designated Eye Point: The designated Pilot Eye Point is located 52 cm from the center of the forward most external view monitor, 61 cm from the left most external view monitor and 24 cm from the ceiling of the simulator enclosure. This point is roughly centered over the pilot's seat when it is adjusted to the forward most position, at a height consistent with the height of the pilot's head.

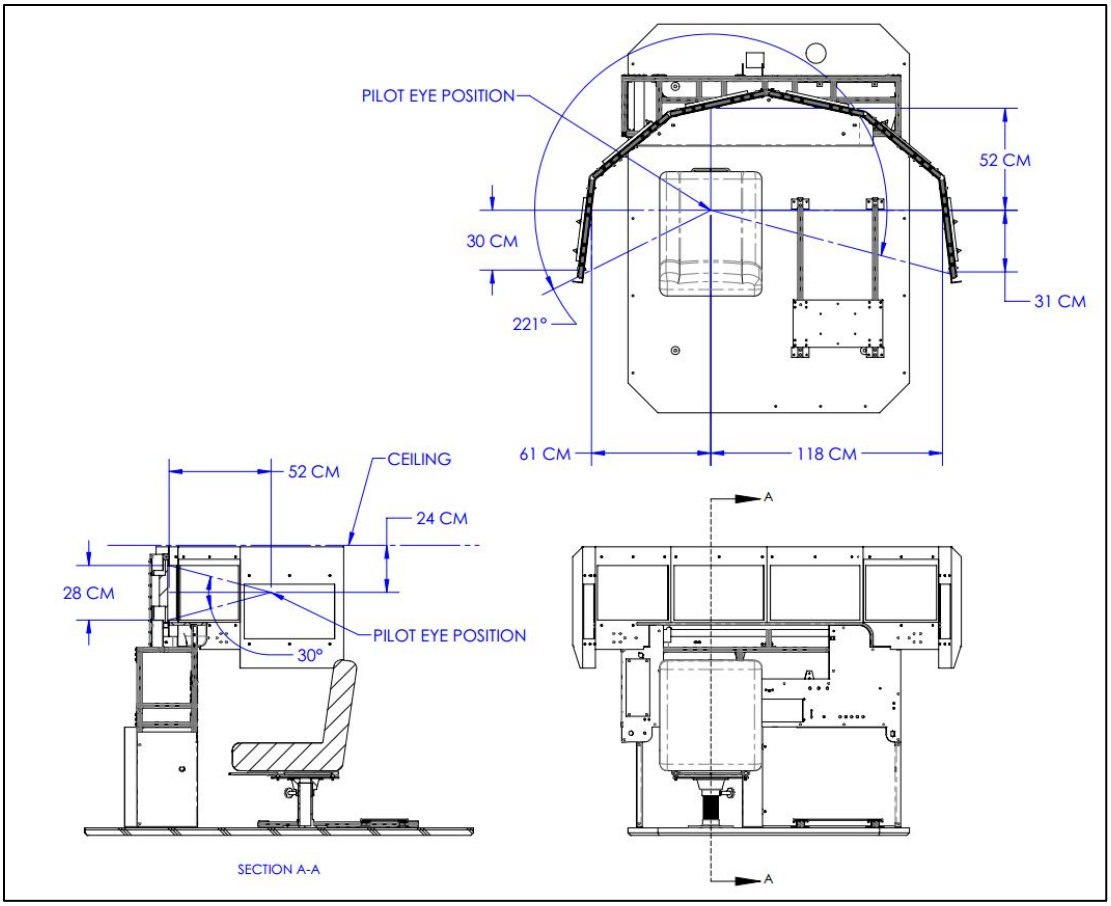


Figure 1: Designated Eye Point Diagram



Figure 6.3: Visual Displays – Type 1

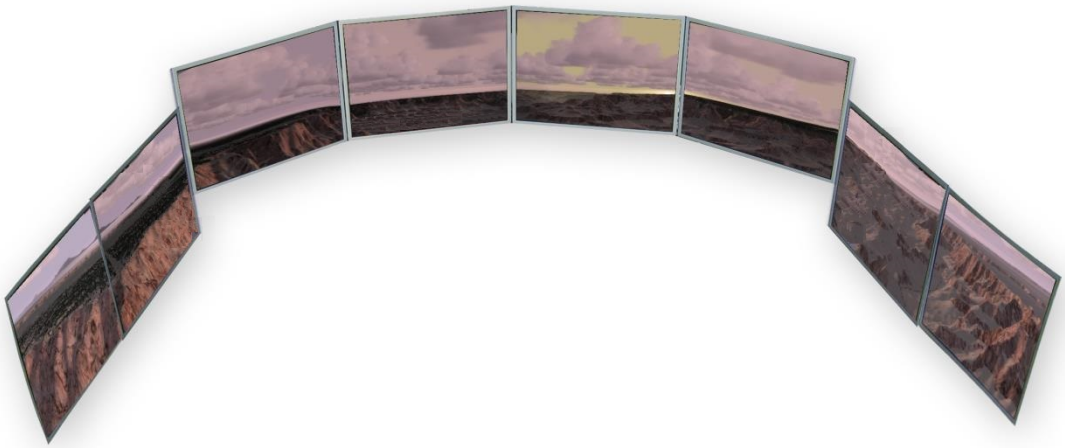


Figure 6.4: Visual Displays – Type 2

SECTION 7: ATD FUNCTIONS & MANEUVERS CHECKLISTS

AVIATION TRAINING DEVICE CHECKLIST

Functions and Maneuvers	Satisfactory? Yes/No
a. Preparation for Flight	
(1) Preflight	Yes
b. Pre-Takeoff	
(1) Engine start	Yes
(2) Brake operation	Yes
c. Takeoff	
(1) **AIRPLANE Takeoff**	
(i) Power plant checks	Yes
(ii) Acceleration characteristics	Yes
(iii) Nose wheel and rudder steering	Yes
(iv) Effect of crosswind	Yes
(v) Instrument	Yes
(vi) Landing gear, wing flap operation	Yes
(2) **HELICOPTER Takeoff**	
(i) Power plant checks	N/A
(ii) From Hover	N/A
(iii) From Ground	N/A
(iv) Vertical	N/A
(v) Running	N/A
d. In-flight Operation	
(1) **AIRPLANE In-flight Operation**	
(i) Climb	Yes
(a) Normal	Yes
(b) One engine inoperative procedures	Yes
(ii) Cruise	
(a) Performance characteristics (speed vs. power)	Yes
(b) Normal and steep turns	Yes
(c) Performance turns	Yes
(d) Approach to stalls, i.e., stall warning (cruise, takeoff/approach, and landing configuration)	Yes
(e) High angle of attack maneuvers (cruise, takeoff/approach, and landing)	Yes
(f) In-flight engine shutdown	Yes
(g) In-flight engine restart	Yes
(iii) Descent	Yes
(a) Normal	Yes
(b) Maximum rate	Yes
(2) **HELICOPTER In-flight Operation**	
(i) Hovering	N/A
(a) Forward	N/A
(b) Rearward	N/A
(c) Sideward	N/A

(d) Turns	N/A
(ii) Climb	N/A
(iii) Cruise	N/A
(a) Performance characteristics	N/A
(b) Turns	N/A
i. Recovery	N/A
ii. Skidding	N/A
iii. Slipping	N/A
(c) In-flight engine shutdown	N/A
(d) In-flight engine restart	N/A
(iv) Descent	N/A
e. Approaches	
(1) Non-precision	Yes
(i) GPS-WAAS	Yes
(ii) GPS-LAAS	Yes
(iii) All engines operating	Yes
(iv) One or more engines inoperative	Yes
(v) Approach procedures (NDB, VOR, DME Arc, LOC/BC, LOC, LDA, SDF, ASR, LNAV/VNAV, GPS, & LPV, and additional types when developed and certified)	Yes
(vi) Missed approach (all engines operational, one or more engines inoperative)	Yes
(2) Precision	Yes
(i) PAR – normal	Yes
(ii) ILS	Yes
(iii) GLS	No
(iv) Effects of crosswind	Yes
(v) With engine(s) inoperative	Yes
(vi) Missed approach	Yes
(a) Normal	Yes
(b) With engine(s) inoperative	Yes
(c) From steep glide slope	Yes
f. Surface Operations (AIRPLANE-Post Landing)	
(1) Landing roll	Yes
(2) Braking operation	Yes
(3) Reverse thrust operation, if applicable	N/A
g. Any Flight Phase	
(1) Aircraft and power plant systems	Yes
(i) Electrical	Yes
(ii) Flaps (**AIRPLANE**)	Yes
(iii) Fuel and oil	Yes
(iv) Landing gear (**AIRPLANE**)	Yes
(2) Flight management and guidance systems	Yes
(i) Automatic pilot	Yes
(ii) Flight director/system displays	Yes
(iii) Navigation systems	Yes
(iv) Stall warning/avoidance (**AIRPLANE**)	Yes
(v) Multi function displays	Yes
(vi) Primary flight displays	Yes

(3) Airborne procedures	Yes
(i) Holding	Yes
(4) Engine shutdown and parking	Yes
(i) Systems operation	Yes
(ii) Parking brake operation (**AIRPLANE**)	Yes

Any changes or modifications to this device which have not been reviewed, evaluated, and approved in writing by General Aviation and Commercial Division, AFS-800 may terminate FAA approval of this aviation training device.